Permutation Signs, Determinants, and Exterior Powers

N-20220725

An $n \times n$ determinant has $n!$ terms in its usual expansion. Yet Gaussian elimination computes it with far fewer operations. The reason is that all those terms implement one simple rule: alternating multilinear dependence on the columns. This treatment follows the determinant and elimination chapters of MIT 18.06 (Strang), with the exterior-power interpretation as a structural continuation.[1]

§1.1 Inversions determine permutation signs

A permutation $\sigma = (\sigma(1), \dots, \sigma(n))$ has an inversion at $i < j$ when $\sigma(i) > \sigma(j)$. Put

$$\operatorname{inv}(\sigma) = \#\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}, \qquad \operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}.$$

For the original example $\sigma = 426315$, the numbers of smaller entries to the right of each position are $3, 1, 3, 1, 0, 0$. Hence

$$\operatorname{inv}(426315) = 3 + 1 + 3 + 1 = 8, \qquad \operatorname{sgn}(\sigma) = +1.$$

Swapping two adjacent positions changes the inversion count by exactly one. Every permutation can be reduced to the identity by adjacent swaps (for example, bubble sort), and consequently its sign is the product of -1 over the swaps. In particular, composing permutations multiplies signs:

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Although the number of swaps in a reduction is not unique, its parity is fixed by the inversion count.

§1.2 Why the Leibniz formula must alternate

For $A = (a_{ij})$, define

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

The terms select one matrix entry in each row and each column. If two columns are interchanged, reindexing permutations by the corresponding transposition pairs the same products with opposite signs; hence the determinant changes sign. It is also linear in each column separately.

A useful characterization is that there is exactly one alternating multilinear function $D(v_1, \dots, v_n)$ with $D(e_1, \dots, e_n) = 1$. Indeed, expand every $v_j = \sum_i a_{ij} e_i$ by multilinearity. Terms with repeated basis vectors vanish; the remaining terms are indexed by permutations and have the signs above.

This argument derives the $n!$-term formula from a short list of geometric axioms, rather than memorizing the formula and reading its consequences backward.

[1] https://web.mit.edu/18.06/www/index.shtml

§1.3 Elementary operations and triangular determinants

Alternation gives $D(\dots, v_i, \dots, v_i, \dots) = 0$. Therefore

$$\det(\dots, v_j + cv_i, \dots) = \det(\dots, v_j, \dots) \quad (i \neq j).$$

Swapping columns changes sign and multiplying one column by c multiplies the determinant by c. Exactly analogous rules hold for rows, since $\det A = \det A^T$.

For a triangular matrix U, the only nonzero term in the Leibniz formula is the identity permutation, so

$$\det U = \prod_{i=1}^{n} U_{ii}.$$

Row reduction is thus a determinant algorithm: record the effects of row exchanges and scalings, and finish by multiplying diagonal entries.

§1.4 A rank-one matrix calculation

Consider the $n \times n$ matrix

$$A = \begin{pmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{pmatrix} = (x-a)I + aJ,$$

where $J = \mathbf{1}\mathbf{1}^T$ has rank one. On the line spanned by $\mathbf{1} = (1, \dots, 1)^T$, J acts by multiplication by n. On the $(n-1)$-dimensional subspace

$$W = \{v : v_1 + \cdots + v_n = 0\},$$

it acts by zero. Since the direct sum of these spaces is $\mathbb{R}^n$,

$$\boxed{\det A = (x + (n-1)a)(x-a)^{n-1}.}$$

For $n = 3$, $x = 2$, $a = 1$, this predicts $\det A = 4$. Direct row elimination on $\left(\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right)$ gives the same result. The symmetry decomposition explains the factorization for every n.

§1.5 Exterior powers remove the choice of coordinates

Let V be n-dimensional. The top exterior power $\bigwedge^n V$ is a one-dimensional vector space. A linear map $A : V \to V$ induces

$$\bigwedge^n A : \bigwedge^n V \longrightarrow \bigwedge^n V, \qquad v_1 \wedge \cdots \wedge v_n \mapsto Av_1 \wedge \cdots \wedge Av_n.$$

An endomorphism of a one-dimensional space is multiplication by a scalar; this scalar is $\det A$. In a basis $e_1, \dots, e_n$,

$$Ae_1 \wedge \cdots \wedge Ae_n = (\det A)\, e_1 \wedge \cdots \wedge e_n.$$

Now multiplicativity has a direct proof: functoriality of the exterior power gives

$$\bigwedge^n (AB) = (\bigwedge^n A)(\bigwedge^n B), \qquad \det(AB) = \det A \det B.$$

We do not need to manipulate two nested sums over permutations.

§1.6 Minors are the matrix of an exterior-power map

For $A : V \to W$ with bases e_i and f_j, the induced map $\bigwedge^k A$ has matrix entries

$$(\bigwedge^k A)_{J,I} = \det A_{J,I},$$

the $k \times k$ minors. This follows by expanding

$$(Ae_{i_1}) \wedge \cdots \wedge (Ae_{i_k})$$

and collecting coefficients of $f_{j_1} \wedge \cdots \wedge f_{j_k}$. Therefore

$$\operatorname{rank} A < k \quad \Longleftrightarrow \quad \bigwedge^k A = 0 \quad \Longleftrightarrow \quad \text{all } k \times k \text{ minors vanish.}$$

It is possible to recognize a rank condition directly from polynomial equations in the matrix entries. This is why minors reappear in algebraic geometry.

§1.7 Cramer's rule is alternating multilinearity again

If A is invertible with columns $a_1, \ldots, a_n$, write

$$b = \sum_{j=1}^{n} x_j a_j.$$

Let $A_i(b)$ be obtained by replacing column i of A by b. Multilinearity yields

$$\det A_i(b) = \sum_j x_j \det(a_1, \ldots, a_{i-1}, a_j, a_{i+1}, \ldots, a_n).$$

All terms with $j \neq i$ have repeated columns and vanish, so $\det A_i(b) = x_i \det A$. Thus

$$x_i = \frac{\det A_i(b)}{\det A}.$$

This proof explains the volume-ratio interpretation precisely, with $\det A \neq 0$ essential. The Leibniz sum, row reduction, minors, exterior powers and Cramer's rule are not five independent pieces of determinant folklore; each is forced by the same alternating multilinear operation.